

Hessian-difference transport, canonical safe packet, and Doob bracket

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-077 closed complete fresh-Gaussian Doob packets in signed expectation and the payload-comparable part of the R-076 shifted resonance. It left a coefficient-dominant form $T_{>}$ and described a complete signed companion packet $\mathfrak{B}_J^>$ by listing the terms which it should contain.

This note makes two corrections before another inequality is attempted. First, the cubic Taylor representation used in R-076 is exact but is not the best global coordinate. The same remainder has an exact Hessian-difference representation with explicit payload A^2DA , not A^3DA . This improves the payload-comparable Young remainder from moment 15 to moment 30/7.

Second, the descriptive list in R-077 does not uniquely assign the terminal square, coefficient curvature, lower Wick chaoses, trace, and finite-low boundary to individual Bony channels. Such an assignment is not canonical: R-063 proves equality of its two second-jet parenthesisations only after the complete forest is summed. We therefore define the remaining packet by subtracting proved orientations from the complete endpoint functional and apply the causal projection once to that whole bracket.

The resulting probability algebra is exact for every packet component which admits the predictable-multiplier/current-increment factorisation displayed below. A coefficient predictable before that current increment centers, but a later adapted coefficient leaves its Doob innovation. This identifies a predictable quadratic-covariation mechanism, not an exact bracket representation of the entire nonlinear safe packet. Its unweighted coefficient coordinate obeys one Hilbert square-function bound. The high- U principal orientation is closed by the canonical R-050 enhancement with moment 60/19. The exact full-packet reconstruction and its future-control weighted bound are not closed here: a termwise absolute route has negative Young slack, and an exact bounded adapted fixture shows that a bracket plus its positive square has no automatic sign.

All statements are at finite cutoff and uniformly in that cutoff at a fixed positive production floor. The physical covariance and filtration remain the pinned common real-even sharp cubes. Smooth Littlewood–Paley blocks are used only for analysis. This result does not prove the weighted bracket, controlled-shell one-use, the $q = 10/9$ Nelson moment, an interacting measure, floor or regulator removal, infinite volume, or Sector-A closure.

2 Exact Hessian-difference transport

Suppress generator, spatial-direction, and tensor sums. Put

$$M_0 = M(U), \quad M_1 = M(U + A), \quad D = DM(U)[A], \quad (2.1)$$

$$E = M_1 - M_0 - D = E_2 + E_3, \quad E_2 = \frac{1}{2}D^2M(U)[A, A]. \quad (2.2)$$

R-076 used

$$E_3 = \frac{1}{2} \int_0^1 (1-t)^2 D^3M(U + tA)[A, A, A] dt. \quad (2.3)$$

Integrating the fundamental theorem of calculus in the opposite order gives the exact alternative

$$\boxed{E_3 = \int_0^1 (1-t) \widehat{K}_t[A, A] dt, \quad \widehat{K}_t = D^2 M(U + tA) - D^2 M(U).} \quad (2.4)$$

Indeed,

$$\int_0^1 (1-t) \int_0^t f(s) ds dt = \frac{1}{2} \int_0^1 (1-s)^2 f(s) ds. \quad (2.5)$$

Thus (2.3) and (2.4) are identical at every smooth cutoff. No Taylor order is discarded.

The production frame has bounded derivatives of every fixed order at the fixed positive floor. In particular, (2.4) gives

$$|E_3| \leq 2 \sup_z \|D^2 M(z)\| |A|^2. \quad (2.6)$$

There are also exact tame derivative identities

$$D_A E_3[h] = \{DM(U + A) - DM(U) - D^2 M(U)[A]\}[h], \quad (2.7)$$

$$\begin{aligned} D_U E_3[h] &= DM(U + A)[h] - DM(U)[h] - D^2 M(U)[h, A] \\ &\quad - \frac{1}{2} D^3 M(U)[h, A, A]. \end{aligned} \quad (2.8)$$

The first right side is a first-order remainder of DM and the second is a second-order remainder of DM . Combining their small- A integral forms with the global bounded-derivative forms yields

$$\boxed{|D_A E_3[h]| \leq C_e |A| |h|, \quad |D_U E_3[h]| \leq C_e |A|^2 |h|.} \quad (2.9)$$

Consequently, schematically but with tensorwise bounded coefficients,

$$D_x E_3 = O_e(A^2) DU + O_e(A) DA. \quad (2.10)$$

The earlier cubic formula remains useful near $A = 0$, but a proof which treats its A^3 payload and shifted $D^3 M$ multiplier separately destroys the global quadratic cancellation in (2.4).

Let

$$J = M(U)^T DU. \quad (2.11)$$

The R-076 E_3 - C transport is now

$$\boxed{N_3 = \int_0^1 (1-t) \langle J, Q_{II}[\widehat{K}_t[A, A]]^T DA \rangle dt.} \quad (2.12)$$

This is the coordinate used below.

3 Sharpened quadratic-payload Besov theorem

Let $0 < s < 1$ and $A \in H^2(\mathbb{T}_L^3) \cap L^6$. Assign every monomial in $A^2 DA$ to its largest input Littlewood–Paley index, with a fixed tie rule, and write

$$A^2 DA = \sum_{r \geq -1} H_r. \quad (3.1)$$

Every H_r is supported in a ball of radius $C2^r$, including its high–high-to-low outputs. The same input-maximum proof as in R-076, now with two undifferentiated factors, gives

$$\|H_r\|_{B_{1,1}^s} \lesssim 2^{(1+s)r} \|\Delta_r A\|_2 \|A\|_6^2. \quad (3.2)$$

The fixed torus embeds the intermediate $L^{6/5}$ product into L^1 . Summing and using the R-076 interpolation

$$\|A\|_{B_{2,1}^{1+s}} \lesssim \|A\|_{H^2}^{(1+s)/2} \|A\|_6^{(1-s)/2} \quad (3.3)$$

proves

$$\|A^2 DA\|_{B_{1,1}^s} \lesssim_s \|A\|_{H^2}^{(1+s)/2} \|A\|_6^{(5-s)/2}. \quad (3.4)$$

Let $X = \|A\|_{H^2}^2$, $Y = \|A\|_6^6$, and let R be the $\mathcal{C}^{-s}$ norm of a control-independent reconstructed current. Then

$$R\|A^2 DA\|_{B_{1,1}^s} \lesssim R X^a Y^b, \quad a = \frac{1+s}{4}, \quad b = \frac{5-s}{12}, \quad c = 1 - a - b = \frac{2-s}{6}. \quad (3.5)$$

One global weighted Young inequality gives

$$R X^a Y^b \leq \eta X + \zeta Y + C_s \eta^{-a/c} \zeta^{-b/c} R^{1/c}. \quad (3.6)$$

At the existing $s = 3/5$,

$$(a, b, c) = \left(\frac{2}{5}, \frac{11}{30}, \frac{7}{30} \right), \quad \frac{1}{c} = \frac{30}{7}, \quad (3.7)$$

and hence

$$R X^{2/5} Y^{11/30} \leq \eta X + \zeta Y + C \eta^{-12/7} \zeta^{-11/7} R^{30/7}. \quad (3.8)$$

Every finite moment is available for the R-050 current. Therefore the nonresonant and payload-comparable orientations of (2.12) close with moment 30/7. R-077's moment-15 cubic-coordinate estimate remains correct, but it is superseded for this exact transport by (3.8).

4 Canonical complete-companion packet

Put $G = DU$, $C = DA$ and

$$w = M_0^T G, \quad \Delta w = (D + E_2 + E_3)^T G, \quad c = M_1^T C, \quad (4.1)$$

$$Q_J = G \otimes G - \Gamma_J, \quad F_A = B(U + A) - B(U) - DB(U)[A]. \quad (4.2)$$

The exact covariance-normal first variation is

$$\mathcal{S}_{0,\Gamma} = \frac{1}{2} \langle DB(U)[A], Q_J \rangle + \langle M_0^T G, Q_{\Pi} M_0^T C \rangle. \quad (4.3)$$

Define

$$L = \langle w, Q_{\Pi} D^T C \rangle, \quad N_2 = \langle w, Q_{\Pi} E_2^T C \rangle, \quad (4.4)$$

and retain the complete endpoint bracket

$$\mathcal{P} = \frac{1}{2} \langle F_A, Q_J \rangle + \langle \Delta w, Q_{\Pi} c \rangle + \frac{1}{2} |c|_{Q_{\Pi}}^2. \quad (4.5)$$

R-075 (3.8) and R-076 (2.6) give exactly

$$\Delta V_J^{\text{ren}} = \mathcal{S}_{0,\Gamma} + L + N_2 + N_3 + \mathcal{P}. \quad (4.6)$$

Here $\mathcal{S}_{0,\Gamma}$ contains both R-073 restored first variations and their Wick trace. Also,

$$\mathcal{P} = \langle w, Q_{\Pi} (E_2 + E_3)^T G \rangle + \frac{1}{2} |\Delta w + c|_{Q_{\Pi}}^2 - \frac{1}{2} F_A : \Gamma_J, \quad (4.7)$$

so the terminal square and coefficient curvature occur exactly once.

Use the same smooth analysis partition in (2.12). Split

$$N_3 = N_{3,\text{nr}} + \widehat{T}_\leq + \widehat{T}_>, \quad (4.8)$$

where $N_{3,\text{nr}}$ contains the two nonresonant paraproducts and

$$\widehat{T}_\leq = \int_0^1 (1-t) \sum_r \sum_{\substack{|m-n| \leq L \\ m \leq r+L}} \int \langle (\Delta_m \widehat{K}_t)(\Delta_n J), Q_{\text{II}} H_r \rangle dx dt, \quad (4.9)$$

$$\widehat{T}_> = \int_0^1 (1-t) \sum_r \sum_{\substack{|m-n| \leq L \\ m > r+L}} \int \langle (\Delta_m \widehat{K}_t)(\Delta_n J), Q_{\text{II}} H_r \rangle dx dt. \quad (4.10)$$

Equality belongs to $\widehat{T}_\leq$. Equations (4.8)–(4.10) are a disjoint exhaustive cutoff identity.

The algebraic safe remainder is defined by subtraction from the complete functional:

$$\boxed{\mathfrak{B}_{J,\text{alg}}^> := \Delta V_J^{\text{ren}} - N_{3,\text{nr}} - \widehat{T}_\leq = \mathcal{S}_{0,\Gamma} + L + N_2 + \widehat{T}_> + \mathcal{P}.} \quad (4.11)$$

This definition, unlike the descriptive R-077 list, assigns no individual lower chaos or square fragment to $\widehat{T}_>$. The whole $F_A : Q_J$ term is interpreted only through the complete R-063 P_3/P_1 and $P_4/P_2/\Sigma Q/P_0$ forest.

For a terminal functional $F(A)$ define the complete predictable projection

$$\mathcal{C}_J F := \sum_{k=j_0}^J \mathbb{E}[F(A^{(k)}) - F(A^{(k-1)}) \mid \mathcal{F}_{k-1}]. \quad (4.12)$$

Then

$$\mathbb{E} \mathcal{C}_J F = \mathbb{E}\{F(A^{(J)}) - F(A^{(j_0-1)})\}. \quad (4.13)$$

The canonical causal packet is

$$\boxed{\mathfrak{B}_{J,\text{safe}}^> := \mathcal{C}_J[\mathfrak{B}_{J,\text{alg}}^>].} \quad (4.14)$$

The initial endpoint in (4.13) is $\mathfrak{B}_{J,\text{alg}}^>(A^{(j_0-1)})$. Its ΔV_{ren} component is the R-066 finite-low endpoint; the values $N_{3,\text{nr}}(A^{(j_0-1)})$ and $\widehat{T}_\leq(A^{(j_0-1)})$ are also retained and must be absorbed by the fixed-low estimate. No low-end term is discarded or asserted to vanish. Crucially, $\mathcal{C}_J$ is applied once to the complete bracket. Applying it separately to forest or square fragments would change the packet and can reintroduce the R-063 associator.

The correction of the previous descriptive object is recorded as

AUDIT-2026-07-25-A13-R077-PACKET-DEFINITION. It changes no proved R-077 identity or bound.

5 Exact bilinear innovation-bracket lemma

Let $P_j = \mathbb{E}[\cdot \mid \mathcal{F}_j]$ and $d_j = P_j - P_{j-1}$. Let $Y_j = d_j Y$ be a complete Hilbert-valued current packet, let H be a terminal adapted coefficient, let P be $\mathcal{F}_{j-1}$ -measurable, and let Λ be any deterministic continuous bilinear spatial contraction. Then

$$\boxed{\mathbb{E} \Lambda(P, H, Y_j) = \mathbb{E} \Lambda(P, d_j H, Y_j).} \quad (5.1)$$

Indeed $P_j H = P_{j-1} H + d_j H$, and the predictable term centers against Y_j . Equivalently, for $H_j = P_j H$ and $Y_j = P_j Y$,

$$\boxed{\mathbb{E}_{j-1}[\langle H_j, Y_j \rangle - \langle H_{j-1}, Y_{j-1} \rangle] = \mathbb{E}_{j-1} \langle d_j H, d_j Y \rangle.} \quad (5.2)$$

Thus, for every component admitting the factorisation in (5.1), complete Doob projection replaces the terminal coefficient by its innovation and leaves a predictable quadratic-covariation bracket, not zero and not a Malliavin derivative. Equations (4.14) and (5.1) do not by themselves prove that the terminal square, coefficient curvature, complete Wick forest, trace, and low-end terms in the entire nonlinear safe packet admit one such bracket representation. That exact packet-to-bracket reconstruction remains part of the successor gate.

If the exact companion in one fixed Hilbert metric is $\|d_j H\|^2/2$, ordinary polarization gives

$$\langle d_j H, d_j Y \rangle + \frac{1}{2} \|d_j H\|^2 = \frac{1}{2} \{ \|d_j (H + Y)\|^2 - \|d_j Y\|^2 \}. \quad (5.3)$$

The final negative term is the fresh injection restored by the R-069/R-070 raw-to-Wick trace. In an adapted production metric, curvature and Wick compensators remain; (5.3) supplies no standalone sign.

The Hilbert martingale square function is nevertheless genuinely one-use:

$$\sum_j \mathbb{E} \|d_j H\|^2 = \mathbb{E} \|H - P_{j_0-1} H\|^2 \leq \mathbb{E} \|H\|^2. \quad (5.4)$$

Conditional projections commute with deterministic smooth spatial blocks, so the same statement holds after summing their L_x^2 squares. Since

$$|\widehat{K}_t| \leq t \sup_z \|D^3 M(z)\| |A|, \quad (5.5)$$

one obtains

$$\int_0^1 (1-t) \sum_{j,m} \mathbb{E} \|\Delta_m d_j \widehat{K}_t\|_2^2 dt \lesssim_e \mathbb{E} \|A\|_2^2. \quad (5.6)$$

This controls the unweighted innovation coordinate once. It does not yet control the spatially weighted bracket in (4.14).

6 The high- U principal orientation

First parilinearisation of the Hessian difference gives

$$\widehat{K}_t = T_{C_t} U + t T_{D^3 M(U+tA)} A + \widehat{K}_t^\sharp, \quad C_t = D^3 M(U+tA) - D^3 M(U). \quad (6.1)$$

The rough principal of J is $T_{M(U)^T} DU$. Define the oriented high- U principal by

$$\mathfrak{J}_{U,t} := T_{C_t M(U)^T} (U \circ DU). \quad (6.2)$$

This is an exact declared Bony subterm, not a claim that every commutator has already been assigned. At fixed floor C_t is uniformly bounded, while $M(U)$ has every finite $L^p(\Omega; L^\infty)$ moment. R-050 supplies the canonical enhancement $U \circ DU \in \mathcal{C}^{-\varepsilon}$ in every finite moment. Hence

$$R_U := \sup_{t \in [0,1]} \|\mathfrak{J}_{U,t}\|_{\mathcal{C}^{-\varepsilon}} \in \bigcap_{p < \infty} L^p(\Omega), \quad (6.3)$$

with a norm independent of the chosen control magnitude.

Pairing (6.2) with the quadratic payload and using (3.4) at $s = \varepsilon$ gives

$$|T_{U,\text{pr}}| \lesssim R_U X^{(1+\varepsilon)/4} Y^{(5-\varepsilon)/12}. \quad (6.4)$$

Its slack and moment are

$$c_U = \frac{2-\varepsilon}{6}, \quad p_U = \frac{6}{2-\varepsilon}. \quad (6.5)$$

At $\varepsilon = 1/10$,

$$X^{11/40}Y^{49/120}, \quad c_U = \frac{19}{60}, \quad p_U = \frac{60}{19}, \quad (6.6)$$

and

$$\boxed{|T_{U,\text{pr}}| \leq \eta X + \zeta Y + C\eta^{-33/38}\zeta^{-49/38}R_U^{60/19}}. \quad (6.7)$$

This closes the declared principal subterm. High coefficients inside the commutator and the A -principal in (6.1) remain in the safe packet.

7 Why the remaining absolute routes fail

For an A -high coefficient shell m use the R-073 interpolation

$$\|a_m\|_6 \lesssim N_m^{-\theta} x_m^{\theta/2} y_m^{(1-\theta)/6}. \quad (7.1)$$

For the payload-maximal shell r in A^2DA , use a fraction ϕ of its control gain. With raw current order s the separated kernel is

$$N_m^{s-\theta} N_r^{1-\phi}, \quad m > r + L. \quad (7.2)$$

Summability requires

$$\theta > s, \quad \phi > 1 + s - \theta, \quad q := \theta + \phi > 1 + s. \quad (7.3)$$

Sequence Holder leaves

$$RX^{q/2}Y^{(4-q)/6}. \quad (7.4)$$

Its Young slack is

$$\boxed{c_A = 1 - \frac{q}{2} - \frac{4-q}{6} = \frac{1-q}{3} < -\frac{s}{3}}. \quad (7.5)$$

At $s = 3/5$, $c_A < -1/5$. Thus the Hessian-difference improvement reduces the amplitude degree but does not make a termwise absolute coefficient- A -high route subcritical. Unlike the old cubic coordinate, a direct high-shell Schur remainder now has amplitude degree six, not eight; it is a viable candidate only if terminal-square/trace polarization supplies the required frequency-weighted absorption before Young.

Automatic bracket positivity also fails. Let $\xi \sim N(0, 1)$ and, on a normalised torus, set

$$\begin{aligned} Y_N &= (\xi^2 - 1) \cos Nx, \\ H_N &= e^{-\xi^2} \cos Nx. \end{aligned} \quad (7.6)$$

Then

$$\mathbb{E}e^{-\xi^2} = \frac{1}{\sqrt{3}}, \quad \mathbb{E}\xi^2 e^{-\xi^2} = \frac{1}{3\sqrt{3}}, \quad (7.7)$$

and therefore

$$\boxed{\mathbb{E} \int H_N Y_N \, dx = -\frac{1}{3\sqrt{3}}} \quad (7.8)$$

for every N . The predictable part of H_N centers, so (7.8) equals the innovation bracket with

$$dH_N = (e^{-\xi^2} - 1/\sqrt{3}) \cos Nx. \quad (7.9)$$

Moreover

$$\mathbb{E} \int \left\{ \lambda(dH_N)Y_N + \frac{\lambda^2}{2}|dH_N|^2 \right\} dx = -\frac{\lambda}{3\sqrt{3}} + \frac{\lambda^2}{4} \left(\frac{1}{\sqrt{5}} - \frac{1}{3} \right), \quad (7.10)$$

which is negative for

$$0 < \lambda < \frac{4}{3\sqrt{3}(1/\sqrt{5} - 1/3)} = 6.759734692 \dots \quad (7.11)$$

This is a finite adapted high-high-to-low method counterfixture, not a production counterexample to (4.14). It omits the production curvature, complete forest, restored first variations, trace, and low boundary.

Equations (7.5) and (7.10) are recorded as NG-2026-07-25-A13-AHIGH-ABSOLUTE-AND-AUTOMATIC-BRACKET. They rule out only the stated shortcuts.

8 Exact successor and proof roadmap

After (2.4), (4.14), (5.1), and (6.7), a candidate unresolved stochastic mechanism is the spatially weighted bracket between a future-control coefficient innovation and a complete current packet, together with the high-coefficient commutators left by (6.1). Before it can be used for the whole endpoint, one must derive an exact packet-to-bracket/commutator decomposition retaining the terminal square, curvature, forest, trace, and the complete low-end safe endpoint. If a_k is the strict-past control at shell k , then

$$d_j a_k = 0 \quad (k \leq j), \quad d_j A = \sum_{k>j} d_j a_k. \quad (8.1)$$

Thus only later feedback survives in each bilinear component covered by (5.1). It need not contain two comparable control factors, so the optional R-077 pair-high estimate cannot be invoked.

The next theorem must first prove the exact full-packet decomposition just described and then establish the production-specific bound

$$\boxed{\mathbb{E}\mathfrak{B}_{J,\text{safe}}^{\geq} \geq -\eta\mathbb{E}X - \zeta\mathbb{E}Y - C_{\eta,\zeta,e,j_0},} \quad (8.2)$$

by a weighted innovation-Carleson estimate after, not before, terminal-square polarization and R-063/R-066 trace reconstruction. The unweighted input (5.6), the reduced quadratic amplitude, and every term in the exact low-end safe endpoint in (4.13) must all be retained.

If (8.2) closes, R-075 fixed-cutoff graph recovery extends the regular result to the declared finite-energy class, R-066 completes controlled-shell one-use, and only then may the existing $q = 10/9$ equivalence be used for Nelson synthesis.

9 Failed-route and evidence map

1. **Cubic payload is correct but nonoptimal.** Equations (2.3)–(2.5) show that its shifted D^3M multiplier and base-frozen term recombine into the bounded Hessian difference with payload A^2DA .
2. **A descriptive companion list is not a packet definition.** Individual R-063 parenthesisations differ before the complete forest sum. Definition (4.14) is by subtraction from the complete endpoint and is therefore nonduplicating.
3. **Repeated Doob centering fails for a bilinear packet term.** The exact survivor of the factorised term is $d_j H d_j Y$ in (5.1)–(5.2), not zero. This is not yet a decomposition of the whole packet.
4. **The Doob square function is useful but unweighted.** Equation (5.6) spends the coefficient coordinate once; it does not supply the spatial Carleson weight required by (8.2).
5. **The high- U principal is cheaper than the old current.** The R-050 second level changes the random order from $-3/5$ to $-1/10$ for the declared principal, giving moment $60/19$.

6. **A-high absolute summation remains supercritical.** The exact conflict is (7.3)–(7.5), even after the payload-degree repair.
7. **Bracket plus square is not automatically positive.** The exact Gaussian fixture (7.6)–(7.11) has a nondecaying negative low mode.
8. **Stein and Clark–Ocone are not substitutes.** Differentiating a later feedback control introduces uncontrolled first and second Malliavin derivatives, as already diagnosed by R-066/R-077.

10 Devil’s-advocate review

1. **“Equation (2.4) approximates the cubic remainder.” DISMISSED.** Fubini identity (2.5) proves exact equality at every smooth cutoff.
2. **“The improved moment contradicts R-077.” DISMISSED.** R-077 bounded a different exact decomposition. Its moment-15 theorem remains valid; the reassembled Hessian coordinate is strictly more efficient.
3. **“One may distribute the lower chaoses among convenient Bony channels.” UPHELD as false.** R-063 fixes equality only after the complete forest. Definition (4.14) forbids such redistribution.
4. **“A predictable coefficient and an adapted terminal coefficient center in the same way.” UPHELD as false.** Equation (5.1) leaves the terminal coefficient innovation.
5. **“The square function (5.6) proves (8.2).” UPHELD as an overclaim.** It has no spatial weight and does not yet couple the terminal square, curvature, or trace.
6. **“The generic bilinear lemma (5.1) is already an exact decomposition of the entire safe packet.” UPHELD as false.** The endpoint algebra and the bilinear Doob lemma are separately exact, but no displayed identity yet factors every square, curvature, forest, trace, and low-end term into (5.1). The successor must prove that reconstruction before using the weighted bracket estimate.
7. **“All high- U commutators are closed by (6.7).” UPHELD as false.** Only the explicitly defined principal (6.2) is closed. High coefficients in commutators remain in the safe packet.
8. **“The negative Gaussian fixture refutes the production theorem.” DISMISSED.** It refutes generic centering and positivity only; it deliberately omits the production companions.
9. **“The optional pair-high estimate closes later feedback.” UPHELD as false.** A single later control can depend on an earlier root, so no second comparable explicit input is forced.
10. **“This proves one-use, Nelson, or Sector A.” UPHELD as an overclaim.** The weighted innovation bracket (8.2) remains open and the tier stays T4.

11 Reproduction and independent audit

The primary executable derives both Taylor representations symbolically, checks the tame remainder identities, reconstructs the endpoint packet in a generic matrix fixture, proves the finite-tree innovation identities and square-function equality, derives every rational exponent, and evaluates the Gaussian anti-centering fixture. The non-importing executable uses separate finite differences, Gauss–Hermite quadrature, a different probability tree, independent fraction arithmetic, and opposite complex Fourier carriers.

Run from the repository root:


```
python codes/foundations/  
a13_classii_hessian_difference_safe_packet_doob_bracket_verify.py
```

The integrated verifier pins R-050, R-063, R-066, R-069, R-070, R-073, R-075, R-076, and R-077; checks the source, PDF, manifest, run records, and repository surfaces; and rejects any assertion that (8.2), one-use, Nelson, or Sector A is closed.

The pre-release attribution repair is recorded as AUDIT-2026-07-25-A13-R078-PRE-RELEASE-PACKET-TO-BRACKET-ATTRIBUTION. It changes no Hessian identity, tame bound, exponent ledger, generic Doob identity, or counterfixture.

Result footer

Result ID: A13-CLASSII-HESSIAN-DIFFERENCE-SAFE-PACKET-DOOB-BRACKET-REDUCTION

Precise statement:	The R-076 E3 transport has an exact bounded Hessian-difference representation with A^2 DA payload. Its comparable block requires moment 30/7. Subtraction from the complete endpoint gives one canonical companion packet. Complete causal projection identifies the future-control Doob-innovation-bracket mechanism for factorised bilinear components; its unweighted coefficient square function is one-use. The declared high-U principal closes with moment 60/19.
Scope:	Fixed positive floor; finite cutoff and uniform cutoff estimates; common real-even regulators; strict whole-shell one-shot controls; smooth LP analysis blocks; complete R-063/R-066 companions.
Dependencies:	A1; R-050; R-063; R-066; R-069; R-070; R-073; R-075; R-076; R-077.
Evidence grade:	ANALYTIC, EXACT, EXECUTED.
Reproduction command:	python codes/foundations/ a13_classii_hessian_difference_safe_packet_ doob_bracket_verify.py
Expected output:	Primary 33/33; non-importing independent 24/24; manifest-pinned integrated and aggregate PASS; exit zero.
Falsification gate:	Disagreement of (2.3) and (2.4); failure of the 2/5, 11/30, 7/30, 30/7 or high-U 11/40, 49/120, 19/60, 60/19 ledgers; failure of (4.6), (5.1), or (5.4); disappearance of the negative fixture; any missing forest companion; or evidence/hash drift.
Tier before / after:	T4 / T4; no promotion.
No-overclaim statement:	The exact full nonlinear packet-to-bracket decomposition, its spatially weighted future-control bound, complete packet lower bound, finite-energy extension, controlled-shell one-use, $q=10/9$ Nelson, interacting measure, all removal limits, and Sector-A closure remain open.
Next required action:	Derive the exact full packet-to-bracket/commutator decomposition, retaining every low-end safe term; then prove (8.2) with square, curvature, complete Wick forest, and trace retained before R-075/R-066.